

D469

# Quality, Continuous Improvement, and Lean Six Sigma

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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# How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

|                             |                                           |                        |                                  |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>_____ | [ ] sure [ ] unsure [ ]<br>guess |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|

# Items

1.

A regional airline's baggage team, led by a Black Belt with 15 years' experience, has defined the problem and gathered six weeks of baseline data on lost bags. They are now using a fishbone diagram and regression to find which factors drive losses. Which DMAIC phase are they in?

- A. Measure
- B. Control
- C. Analyze
- D. Improve

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2. [TRANSFER — reversed stem / different variable]

In which DMAIC phase does a team create a control plan and SPC charts to sustain its gains?

- A. Improve
- B. Define
- C. Measure
- D. Control

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

An appliance maker spent \$1.8 million on warranty repairs last year. Its Ohio plant has 900 workers. Which cost-of-quality category are warranty repairs?

- A. Internal failure
- B. External failure
- C. Appraisal
- D. Prevention

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**4.** [TRANSFER — reversed stem / different variable]

Which spending is a prevention cost?

- A.** Inspecting finished units at the end of the line.
- B.** Scrapping defective parts found in the plant.
- C.** Training operators on a mistake-proofing fixture before production starts.
- D.** Handling customer returns.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**5.**

A bottling line's fill-weight chart, reviewed by a quality engineer hired last month, shows every point inside the control limits, but the last eight points in a row are above the center line. The line runs 24 hours a day. What should the engineer conclude?

- A.** The run signals a special cause or process shift; investigate.
- B.** Widen the control limits to match the spec limits.
- C.** Adjust the filler after every point.
- D.** The process is in control because all points are inside the limits.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**6.**

A machining cell makes shafts with spec limits of 24.95 mm to 25.05 mm. The process standard deviation is 0.02 mm. The cell was installed in 2019 by a German supplier. What is Cp?

- A.** 1.20
- B.** Capable, as long as the mean is centered
- C.** About 1.67, capable
- D.** About 0.83, not capable

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**7.** [TRANSFER — reversed stem / different variable]

A process has a tolerance width (USL – LSL) of 0.60 mm. What is the largest standard deviation that still gives Cp = 1.33?

- A.** About 0.80 mm
- B.** 0.10 mm
- C.** About 0.15 mm
- D.** About 0.075 mm

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8.

A claims center processed 2,000 claims last month. Each claim has 5 opportunities for error. Auditors found 60 errors. The center's director was featured in a trade magazine. What is DPMO?

- A. 30,000
- B. 3% defective
- C. 6,000
- D. 60

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

9. [TRANSFER — reversed stem / different variable]

A process runs at 3,000 DPMO. It makes 400,000 units with 4 opportunities each. How many defects are expected?

- A. 4,800
- B. 3,000
- C. 1,200
- D. 300

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

10.

A hotel chain logged 11,000 guest complaints last year: room noise 4,400, slow check-in 3,300, Wi-Fi 1,650, parking 880, and other 770. The chain recently changed its logo. With budget for two projects, which should it pick?

- A. Wi-Fi and parking, since they are smaller and easier.
- B. Room noise and slow check-in.
- C. Spread the budget evenly across all five categories.
- D. A campaign promoting the new logo.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

**11.**

At a clinic, lab results sit in an outbox for an average of 3 hours before a courier takes them to the physician’s office down the hall. The clinic just upgraded its lab analyzers. Which lean waste is the 3-hour delay?

- A.** Inventory
- B.** Waiting
- C.** Transportation
- D.** Overprocessing

|                             |                                           |                       |                                                                                                 |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>____ | <input type="checkbox"/> sure <input type="checkbox"/> unsure <input type="checkbox"/><br>guess |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|

**12.**

Customers say a smartphone maker’s phones last five years before wearing out, though its camera specs trail rivals. The company sponsors a pro soccer team. Which dimension of quality is “last five years before wearing out”?

- A.** Durability
- B.** Reliability
- C.** Perceived quality
- D.** Performance

|                             |                                           |                       |                                                                                                 |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>____ | <input type="checkbox"/> sure <input type="checkbox"/> unsure <input type="checkbox"/><br>guess |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer C Concept 1

A regional airline's baggage team, ~~led by a Black Belt with 15 years' experience~~, has defined the problem and gathered six weeks of baseline data on lost bags. They are now using a fishbone diagram and regression to find which factors drive losses. Which DMAIC phase are they in?

**Target word:** current phase **Decides:** Finding root causes is Analyze.

|   |                   |                                               |
|---|-------------------|-----------------------------------------------|
| A | Adjacent category | Baseline collection is already done.          |
| B | Dropped element   | Skips analysis and improvement entirely.      |
| C | <b>Correct</b>    | Fishbone and regression identify root causes. |
| D | Confusion set     | Solutions come after causes are confirmed.    |

## 2. Answer D Concept 1 • transfer

In which DMAIC phase does a team create a control plan and SPC charts to sustain its gains?

**Target word:** reverse: activity to phase **Decides:** Sustaining gains is Control.

|   |                    |                                                           |
|---|--------------------|-----------------------------------------------------------|
| A | Adjacent category  | Improve pilots and implements solutions.                  |
| B | Inverted operation | The first phase, not the last.                            |
| C | Confusion set      | Measurement data are collected, but not to sustain gains. |
| D | <b>Correct</b>     | Control locks in improvements with plans and monitoring.  |

## 3. Answer B Concept 2

An appliance maker spent \$1.8 million on warranty repairs last year. ~~Its Ohio plant has 900 workers.~~ Which cost-of-quality category are warranty repairs?

**Target word:** category **Decides:** Defects reached customers.

|   |                    |                                                              |
|---|--------------------|--------------------------------------------------------------|
| A | Confusion set      | Internal failures are caught before shipment.                |
| B | <b>Correct</b>     | Costs from defects found after the customer has the product. |
| C | Adjacent category  | Appraisal is inspection and testing.                         |
| D | Inverted operation | Prevention is spent before defects occur.                    |

## 4. Answer C Concept 2 • transfer

Which spending is a prevention cost?

**Target word:** reverse: category to example **Decides:** Prevention stops defects from being made.

|   |                    |                                       |
|---|--------------------|---------------------------------------|
| A | Confusion set      | That is appraisal.                    |
| B | Adjacent category  | That is internal failure.             |
| C | <b>Correct</b>     | Spent to keep defects from happening. |
| D | Inverted operation | That is external failure.             |

## 5. Answer A Concept 3

A bottling line's fill-weight chart, reviewed by a quality engineer ~~hired last month~~, shows every point inside the control limits, but the last eight points in a row are above the center line. ~~The line runs 24 hours a day.~~ What should the engineer conclude?

**Target word:** chart interpretation **Decides:** Eight in a row on one side is a non-random run.

|   |                   |                                                       |
|---|-------------------|-------------------------------------------------------|
| A | <b>Correct</b>    | Run rules detect shifts even inside the limits.       |
| B | Confusion set     | Control limits come from the process, not from specs. |
| C | Adjacent category | Tampering adds variation; find the cause instead.     |
| D | Dropped element   | Ignores the run rules.                                |

## 6. Answer D Concept 4

A machining cell makes shafts with spec limits of 24.95 mm to 25.05 mm. The process standard deviation is 0.02 mm. ~~The cell was installed in 2019 by a German supplier.~~ What is Cp?

**Target word:** Cp **Decides:**  $0.10 / (6 \times 0.02)$ .

|   |                    |                                                            |
|---|--------------------|------------------------------------------------------------|
| A | Inverted operation | $0.12 / 0.10$ divides the process spread by the tolerance. |
| B | Confusion set      | Centering matters for Cpk; Cp is under 1.0 regardless.     |
| C | Dropped element    | $0.10 / 0.06$ uses $3\sigma$ instead of $6\sigma$ .        |
| D | <b>Correct</b>     | $0.10 / 0.12 \approx 0.83$ , below 1.0.                    |

## 7. Answer D Concept 4 · transfer

A process has a tolerance width (USL – LSL) of 0.60 mm. What is the largest standard deviation that still gives Cp = 1.33?

**Target word:** solve for sigma **Decides:**  $\sigma = 0.60 / (6 \times 1.33)$ .

|   |                    |                                       |
|---|--------------------|---------------------------------------|
| A | Inverted operation | Multiplies 0.60 by 1.33.              |
| B | Adjacent category  | That gives Cp = 1.00.                 |
| C | Dropped element    | Uses $3\sigma$ instead of $6\sigma$ . |
| D | <b>Correct</b>     | $0.60 / 8 \approx 0.075$ .            |

## 8. Answer C Concept 5

A claims center processed 2,000 claims last month. Each claim has 5 opportunities for error. Auditors found 60 errors. ~~The center's director was featured in a trade magazine.~~ What is DPMO?

**Target word:** DPMO **Decides:**  $60 / (2,000 \times 5) \times 1,000,000$ .

|   |                   |                                           |
|---|-------------------|-------------------------------------------|
| A | Dropped element   | Leaves out the 5 opportunities per claim. |
| B | Confusion set     | A per-unit defect rate, not DPMO.         |
| C | <b>Correct</b>    | $60 / 10,000 \times 1,000,000 = 6,000$ .  |
| D | Wrong stem number | The raw defect count.                     |

### 9. Answer A Concept 5 • transfer

A process runs at 3,000 DPMO. It makes 400,000 units with 4 opportunities each. How many defects are expected?

**Target word:** solve for defects **Decides:** 1,600,000 opportunities  $\times$  0.003.

|   |                    |                                                       |
|---|--------------------|-------------------------------------------------------|
| A | <b>Correct</b>     | $400,000 \times 4 \times 3,000 / 1,000,000 = 4,800$ . |
| B | Wrong stem number  | The DPMO rate itself.                                 |
| C | Dropped element    | Leaves out opportunities per unit.                    |
| D | Inverted operation | Divides by opportunities instead of multiplying.      |

### 10. Answer B Concept 6

A hotel chain logged 11,000 guest complaints last year: room noise 4,400, slow check-in 3,300, Wi-Fi 1,650, parking 880, and other 770. ~~The chain recently changed its logo.~~ With budget for two projects, which should it pick?

**Target word:** priority **Decides:** Noise and check-in are 70% of complaints.

|   |                    |                                            |
|---|--------------------|--------------------------------------------|
| A | Inverted operation | Picks the trivial many.                    |
| B | <b>Correct</b>     | 7,700 of 11,000 complaints: the vital few. |
| C | Dropped element    | Drops the prioritization Pareto provides.  |
| D | Camouflage         | Unrelated to any complaint category.       |

### 11. Answer B Concept 7

At a clinic, lab results sit in an outbox for an average of 3 hours before a courier takes them to the physician's office down the hall. ~~The clinic just upgraded its lab analyzers.~~ Which lean waste is the 3-hour delay?

**Target word:** waste type **Decides:** Results sit idle between steps.

|   |                   |                                                                        |
|---|-------------------|------------------------------------------------------------------------|
| A | Confusion set     | Refers to excess stock or WIP, not delay time.                         |
| B | <b>Correct</b>    | Idle time between process steps.                                       |
| C | Adjacent category | The courier's trip is transport; the 3 hours in the outbox is waiting. |
| D | Camouflage        | Chases the analyzer upgrade, which is not the delay.                   |

### 12. Answer A Concept 8

Customers say a smartphone maker's phones last five years before wearing out, though its camera specs trail rivals. ~~The company sponsors a pro soccer team.~~ Which dimension of quality is "last five years before wearing out"?

**Target word:** quality dimension **Decides:** Useful life before wearing out.

|   |                   |                                                |
|---|-------------------|------------------------------------------------|
| A | <b>Correct</b>    | How long the product lasts.                    |
| B | Confusion set     | How often it fails within a given period.      |
| C | Camouflage        | Brand image, as from the sponsorship.          |
| D | Adjacent category | Primary operating traits such as camera specs. |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 7             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 2             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 8             |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 7             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 0             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 9             |             |
| Camouflage         | Answers a stem fact that was never the question.             | 3             |             |

## Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                  | Rule                                                                                                                         | Trap                                                                             | Tell                                               |
|---|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------|
| 1 | <b>DMAIC phases</b>                      | Define the problem, Measure the baseline, Analyze root causes, Improve with solutions, Control to sustain.                   | Placing root-cause tools in Measure or control plans in Improve.                 | A list of what the team has done and is doing now. |
| 2 | <b>Cost of quality categories</b>        | Prevention and appraisal are spent to avoid defects; internal failure is found before shipment, external failure after.      | Calling warranty costs internal failure.                                         | Whether the defect reached the customer.           |
| 3 | <b>Control charts and special causes</b> | Points outside limits or non-random runs (such as 7-8 in a row on one side) signal special causes.                           | Thinking “all points inside the limits” always means in control.                 | A run of points on one side of the center line.    |
| 4 | <b>Process capability (Cp)</b>           | $C_p = (USL - LSL) / 6\sigma$ ; below 1.0 the process cannot fit inside the spec.                                            | Dividing by $3\sigma$ or by $\sigma$ .                                           | Spec limits and a standard deviation.              |
| 5 | <b>DPMO</b>                              | $DPMO = \text{defects} / (\text{units} \times \text{opportunities per unit}) \times 1,000,000$ .                             | Leaving out opportunities per unit.                                              | Units, opportunities per unit, and defects.        |
| 6 | <b>Pareto analysis</b>                   | Focus effort on the vital few categories that account for most of the problem.                                               | Spreading effort evenly or choosing the easiest fixes.                           | Counts by category with a limited budget.          |
| 7 | <b>Lean wastes</b>                       | Waiting is idle time between steps; transportation is moving items; inventory is excess stock.                               | Confusing waiting with transportation.                                           | Items sitting still vs. items being moved.         |
| 8 | <b>Dimensions of quality</b>             | Performance, features, reliability, conformance, durability, serviceability, aesthetics, and perceived quality are distinct. | Mixing durability (useful life) with reliability (failure-free within a period). | How long a product lasts vs. how often it fails.   |

**Course scope (public description):** Quality, Continuous Improvement, and Lean Six Sigma examines how organizations can measure and improve the quality of products, goods, and services. Students consider different dimensions and characteristics of quality and are introduced to a variety of strategies and tools that are used to manage quality and measure performance. This course explores how to apply continuous improvement processes by analyzing the Lean Six Sigma DMAIC (define, measure, analyze, improve, and control) method. C955: Applied Probability and Statistics is a prerequisite for this course.